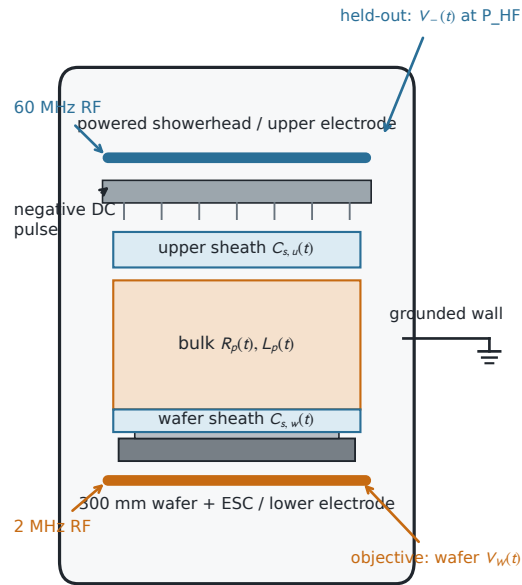


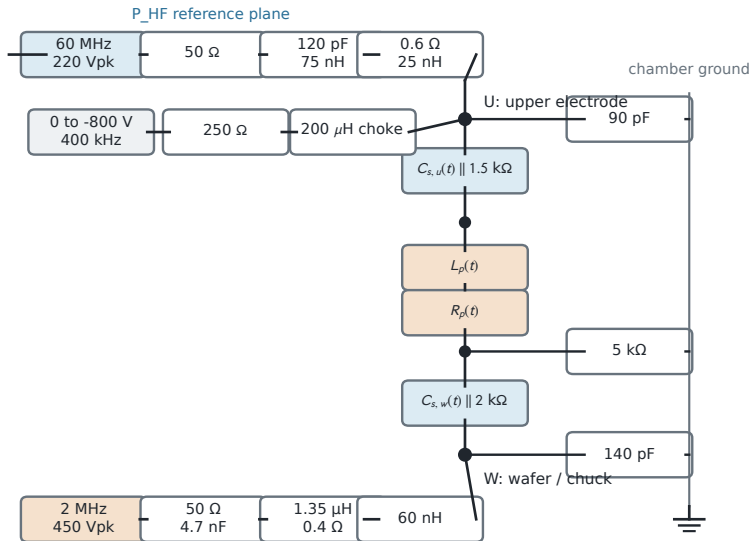
Dual-frequency + DC-superposed etch CCP with dynamic plasma impedance

Hardware and circuit share the same vertical order; the matching, feed, choke, blocking and stray elements remain explicit.

(a) 300 mm etch-CCP problem and outputs

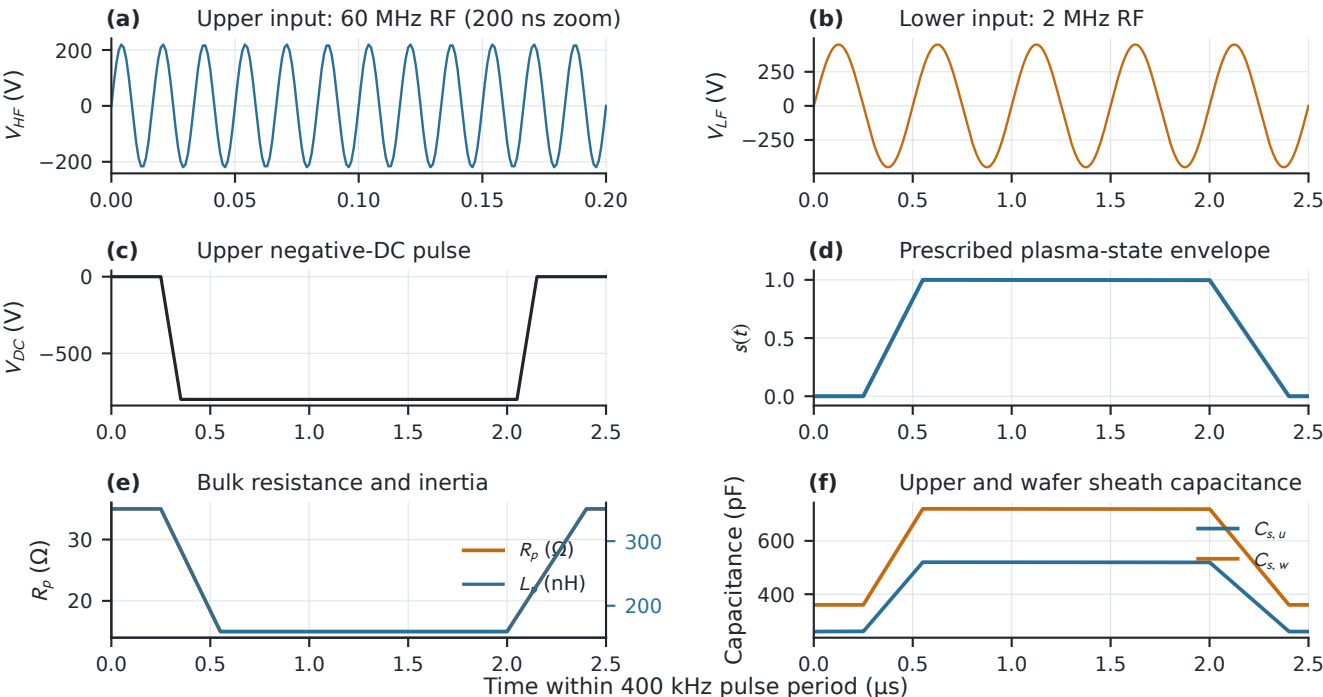


(b) Top-to-bottom external + plasma equivalent circuit



Known excitations and prescribed dynamic plasma elements

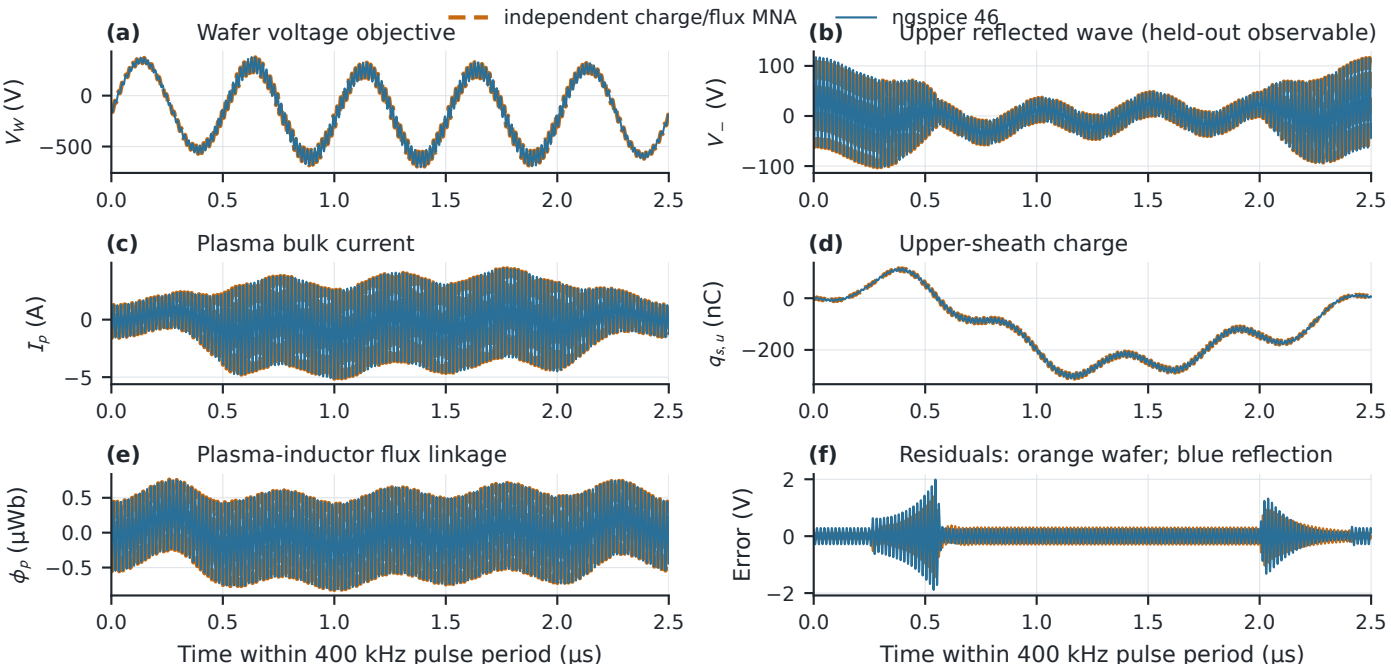
The envelope moves R_p : 35→15 Ω , L_p : 350→160 nH, $C_{s,u}$: 260→520 pF and $C_{s,w}$: 360→720 pF.



Constitutive laws used by both solvers: $q_s = C_s(t)v_s$ and $\phi_p = L_p(t)i_p$; therefore $i_s = dq_s/dt$ and $v_L = d\phi_p/dt$.

Forward validity: independent state equations and ngspice agree

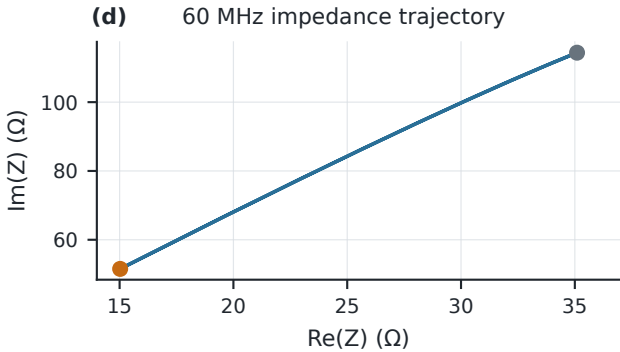
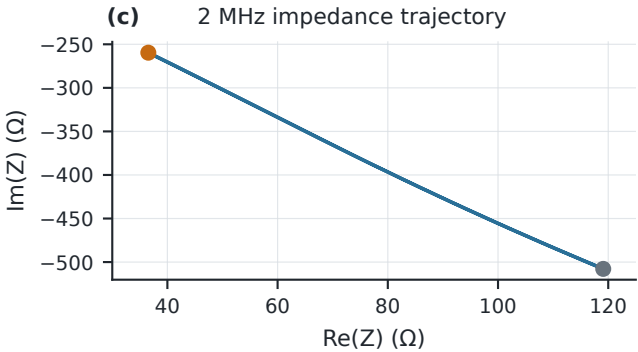
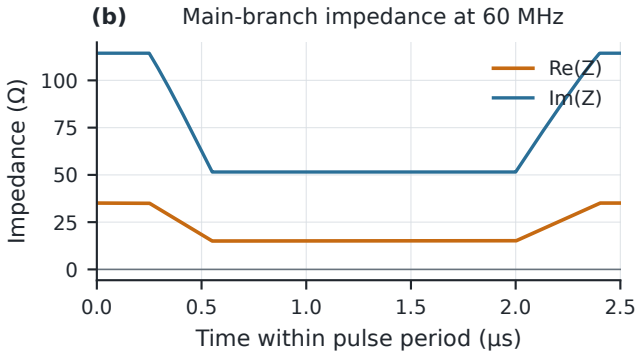
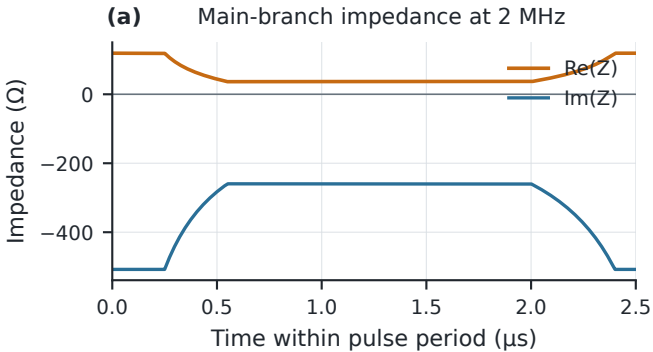
Voltage, current, charge and flux are compared directly over one complete 400 kHz pulse period.



Wafer RMSE=0.287 V (nRMSE=8.132e-04); reflection RMSE=0.364 V (nRMSE=8.784e-03).
Bulk-current RMSE=1.403e-02 A; all curves use archived simulation artifacts.

Electrical interpretation of the time-varying plasma branch

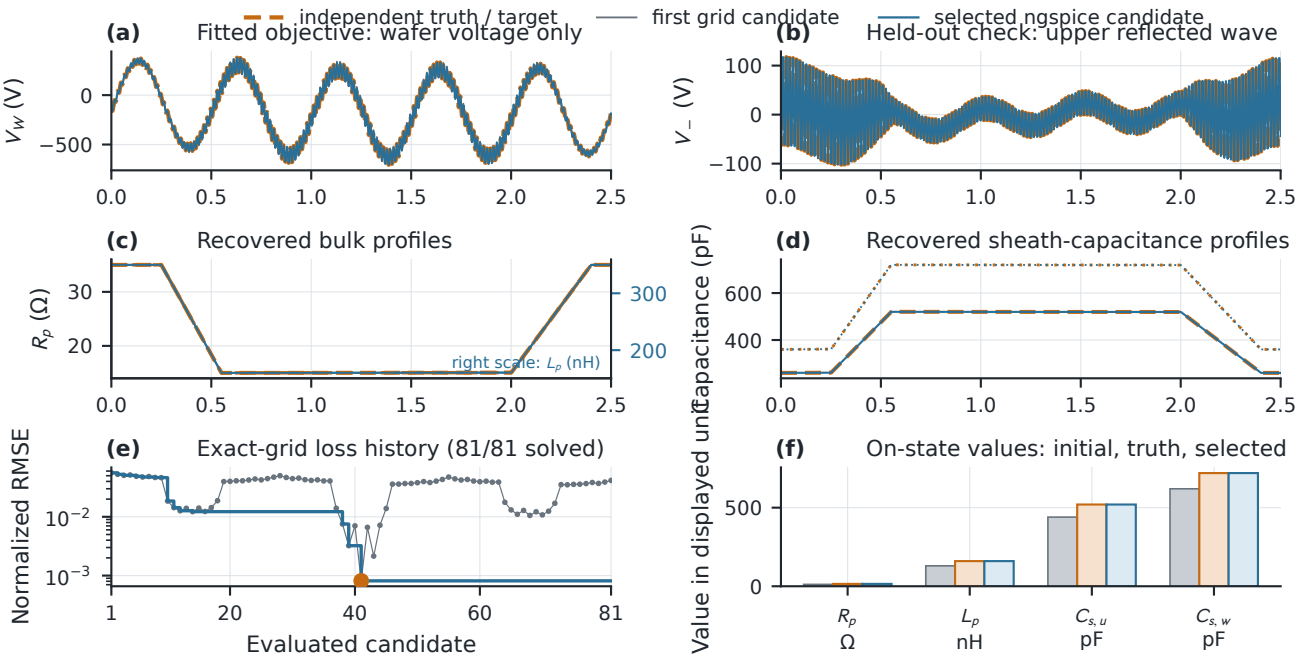
$Z_{main} = Z_{s,u} + R_p + j\omega L_p + Z_{s,w}$ with $Z_s = (1/R_s + j\omega C_s)^{-1}$; external matching is excluded here.



Gray marker: pulse start (low-density state); orange marker: on-state.
Frozen-time impedance explains RF loading but is not a kinetic plasma solution.

Inverse validity: one measured waveform recovers four bounded on-state values

Only wafer voltage contributes to loss; upper reflection is retained as a genuinely unused circuit-level validation output.



Selected: $R_p=15\ \Omega$, $L_p=160\ \text{nH}$, $C_{su}=520\ \text{pF}$, $C_{sw}=720\ \text{pF}$. Objective nRMSE=8.132e-04; held-out nRMSE=8.784e-03.